5. Getting Started with Statistics

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7/12/20

Outline

1. Introduction to Descriptive Statistics
2. Summarizing Data
3. Candidate Distributions
4. Introduction to Estimation
5. Unbiased Estimation
6. Mean Squared Error
7. Maximum Likelihood Estimation
8. Trickier MLE Examples
9. Invariance Property of MLEs
10. Method of Moments Estimation
11. Sampling Distributions

Lesson 5.1 — Introduction to Descriptive Statistics

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- The comparison of many populations.

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Idea (Election polling example): We can't poll every single voter. Thus, we take a **sample** of data from the **population** of voters, and try to make a reasonable conclusion based on that sample.

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- **Categorical variables:** These data are not typically numerical. What's your favorite TV show during a certain time slot?

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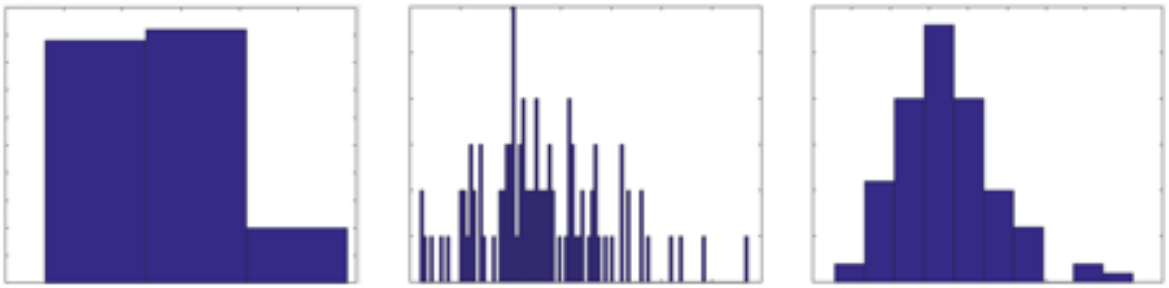

Not enuf cells | Too many | Just right!

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23	62	91	83	82	64	73	94	94	52
67	11	87	99	37	62	40	33	80	83
99	90	18	73	68	75	75	90	36	55

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9	9944100
8	73320
7	5533
6	87422
5	52
4	0
3	763
2	3
1	81

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Range	Freq.	Cumul. Freq.	Proportion of observations so far
0–20	2	2	2/30
21–40	5	7	7/30
41–60	2	9	9/30
61–80	10	19	19/30
81–100	11	30	1

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Remark: Before you take any observations, $\bar{X}$ and S^2 must be regarded as *random variables*.

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Sample Range: $\max_i X_i - \min_i X_i$.

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Then $\bar{X} = (2 \cdot 1 + 1 \cdot 2 + \cdots + 3 \cdot 6)/10 = 3.7$, and $S^2 = 3.789$. $\square$

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- Univariate/multivariate?

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- Univariate/multivariate?
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- What if we do not have much/any data — can we at least guess at a good distribution?

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Examples of parameters: μ, σ^2.

Let $X_1, \ldots, X_n$ be iid RV's and let $T(\mathbf{X}) \equiv T(X_1, \ldots, X_n)$ be a statistic based on the X_i's.

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Remark: S is *not* unbiased for the standard deviation σ.

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If they're all unbiased, which one's the best?

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Lastly, let's look at…

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Therefore, let's look at some other properties an estimator can have.

For instance, consider the *variance* of an estimator.

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Thus, both Y_1 and Y_2 are unbiased, but Y_2 has *much lower variance* than Y_1. We can break the "unbiasedness tie" by choosing Y_2. $\square$

Outline

1. Introduction to Descriptive Statistics
2. Summarizing Data
3. Candidate Distributions
4. Introduction to Estimation
5. Unbiased Estimation
6. Mean Squared Error
7. Maximum Likelihood Estimation
8. Trickier MLE Examples
9. Invariance Property of MLEs
10. Method of Moments Estimation
11. Sampling Distributions

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So MSE $=$ Bias^2 $+$ Var, and thus combines the bias and variance of an estimator. $\square$

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Outline

1. Introduction to Descriptive Statistics
2. Summarizing Data
3. Candidate Distributions
4. Introduction to Estimation
5. Unbiased Estimation
6. Mean Squared Error
7. Maximum Likelihood Estimation
8. Trickier MLE Examples
9. Invariance Property of MLEs
10. Method of Moments Estimation
11. Sampling Distributions

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Remark: We can very informally regard the MLE as the "most likely" estimate of θ.

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- At the end, we make all of the little x_i's into big X_i's to indicate that this is a random variable.
- Just to be careful, you "probably" ought to do a second-derivative test.

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$$\hat{p} \;=\; \bar{X}.$$

This implies that

$$\ell\mathrm{n}(L(p)) \;=\; \sum_{i=1}^{n} x_i\,\ell\mathrm{n}(p) + \Big(n - \sum_{i=1}^{n} x_i\Big)\ell\mathrm{n}(1-p)$$

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This makes sense since $\mathrm{E}[X] = p$. $\square$

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where $\Psi(r) \equiv \Gamma'(r)/\Gamma(r)$ is the **digamma function.**

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$$\Gamma'(r) \;\doteq\; \frac{\Gamma(r+h) - \Gamma(r)}{h} \quad \text{(for any small } h \text{ of your choosing).} \quad \square$$

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Remark: We used very little calculus in this example!

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Remark: The proof of the Invariance Property is "easy" when $h(\cdot)$ is a one-to-one function.

Lesson 5.9 — Invariance Property of MLEs

We can get MLEs of functions of parameters almost for free!

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Remark: The proof of the Invariance Property is "easy" when $h(\cdot)$ is a one-to-one function. It's not so easy — but still generally true — when $h(\cdot)$ is nastier.

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This kind of thing is used all of the time in the actuarial sciences. $\square$

Outline

1. Introduction to Descriptive Statistics
2. Summarizing Data
3. Candidate Distributions
4. Introduction to Estimation
5. Unbiased Estimation
6. Mean Squared Error
7. Maximum Likelihood Estimation
8. Trickier MLE Examples
9. Invariance Property of MLEs
10. Method of Moments Estimation
11. Sampling Distributions

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Remark: You should always love your MoM!

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Let's finish up with a less-trivial example. . . .

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To finish up, you can plug back into Equation (1) to get the MoM estimator for a.

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Outline

1. Introduction to Descriptive Statistics
2. Summarizing Data
3. Candidate Distributions
4. Introduction to Estimation
5. Unbiased Estimation
6. Mean Squared Error
7. Maximum Likelihood Estimation
8. Trickier MLE Examples
9. Invariance Property of MLEs
10. Method of Moments Estimation
11. Sampling Distributions

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In reality, df corresponds to the number of dimensions of a certain space (not covered in this course)!

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For large k, the $\chi^2(k)$ is approximately normal (by the CLT).

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"Student" is the pseudonym of the guy (William Gossett) who first derived it. Gossett was a statistician at the Guinness Brewery.

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t distribution is a special case — can you figure out which?

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